FSD^3: Few-Shot SAR Object Detection With Dynamic Perception RPN and Category Knowledge Decoder

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***Abstract*— Limited by observational constraints and noncooperative targets, synthetic aperture radar (SAR) images often suffer from extreme scarcity of training samples for certain classes, which severely degrades the detection performance and leads to the challenging issue of few-shot SAR object detection. To address this, we propose a Few-Shot SAR object Detection model with Dynamic perception region proposal network (RPN) and category knowledge Decoder, dubbed as FSD^3. Specifically, a dynamic perception RPN is employed to infuse global semantics of support classes into local features of query feature maps, thereby enhancing the recall rate of novel-class targets under few-shot conditions. Subsequently, a category knowledge decoder is designed, which automatically decodes aggregated features by utilizing query features as prompts and effectively mitigates the feature-label mismatch phenomenon. Finally, a dual-constrained detection head (DCDH) is proposed, which leverages the large margin cosine loss and center calibration loss (CCL) to explicitly achieve interclass separation and intraclass compactness, thereby achieving high-precision and robust classification in few-shot scenarios. Experimental results on the public SAR-AIRCraft-1.0 dataset and our custom MSTAR-Fewshot dataset have demonstrated that the proposed model achieves superior few-shot object detection performance across various experimental configurations.**

***Index Terms*— Dynamic network, few-shot object detection, metalearning, object detection, synthetic aperture radar (SAR).**

I. Introduction

AS AN active microwave imaging sensor, synthetic aperture radar (SAR) plays a vital role in battlefield reconnaissance and situational awareness due to its all-day, all-weather, long operating range, and high-resolution imaging capabilities [1]. In recent years, the rapid development of SAR systems and imaging technologies [2], [3], [4], [5], [6] has enabled the acquisition of abundant high-resolution SAR imagery, thereby laying a solid foundation for SAR image interpretation tasks such as object detection.

With the rapid advancement of deep learning technology [7], [8], [9], researchers have investigated deep learning-based SAR object detection methods [10], [11], [12], [13] by integrating SAR image characteristics with generic object detection approaches [14], [15], [16], [17], [18]. Leveraging the powerful feature extraction capabilities and end-to-end framework of deep networks, these methods transform the traditional three-stage SAR object detection and recognition pipeline [19], [20], [21] into a unified detection network. Such innovation not only enhances the detection accuracy but also improves the processing efficiency, thereby becoming the mainstream for SAR object detection.

Constrained by limited observation conditions and noncooperative targets, however, the training data for specific classes in the SAR imagery may become extremely scarce [23], [24]. Moreover, SAR image annotation relies heavily on domain expertise, thereby making data labeling time-consuming and labor-intensive. The above factors render SAR object detection in few-shot issue, where the lack of training data for specific classes severely impairs or even entirely undermines the detection capabilities.

Current few-shot object detection methods mainly focus on the optical imagery and generally fall into two categories: transfer learning-based approaches and metalearning-based approaches [25], [26]. The former gains the detection capability of novel classes through a two-stage training framework, i.e., the model undergoes base training with abundant base class data followed by fine-tuning with limited novel class samples while maintaining an almost unchanged architecture [27], [28], [29], [30], [31], [32], [33]. The latter employs weight-shared Siamese feature extraction networks to process query and support samples independently and then aggregates their features through carefully designed operations to accomplish object detection tasks [34], [35], [36], [37], [38], [39], [40], [41]. Due to the explicit and efficient utilization of novel-class training samples during inference, metalearning-based few-shot object detection has attracted intensive research interests in recent years.

However, the existing few-shot object detection methods cannot be directly applied to SAR object detection since the inherent characteristics of the SAR imagery are not given full consideration, and the following issues are not tackled.

1) Foreground–Background Confusion: Due to the special imaging mechanism, the objects in SAR images are composed of a series of discrete scattering centers, as illustrated in Fig. 1(a). The structural discontinuities, along with background clutter and speckle noise, are prone to causing confusion between the foreground and the background. This issue is

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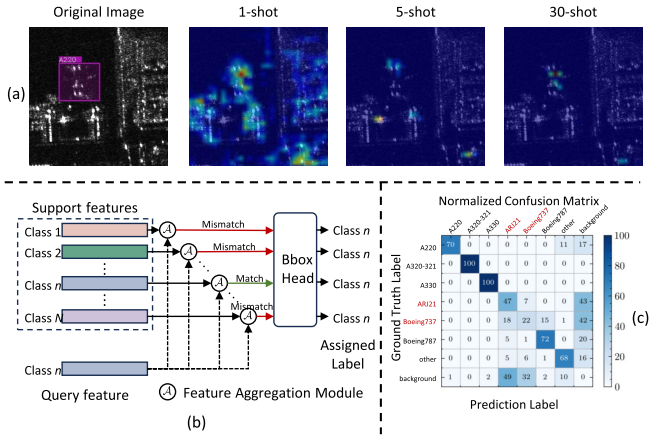


Fig. 1. Key issues encountered in few-shot SAR object detection. (a) Foreground–background confusion. Due to the presence of speckle noise and background clutter interference, the model struggles to effectively separate foreground and background regions under extreme training data scarcity, as illustrated by the Grad-CAM [22] attention visualization maps under 1-, 5-, and 30-shot settings. (b) Feature-label misalignment. The current aggregation approaches ignore the impact of support features during label assignment, resulting in a mismatch between aggregated features and their corresponding labels. (c) Confusion matrix of Meta RCNN on the test set, with novel classes highlighted in red. The small number of training samples leads to confusion between different novel classes, as well as confusion between novel classes and their corresponding similar base classes.

particularly prominent under the condition of limited training samples.

2) Feature-Label Misalignment: Few-shot object detectors based on metalearning first aggregate features extracted from the support branch and the query branch in a specific way and then feed them into the classifier and the bounding box regressor to obtain the detection results [34], [35]. However, there is a mismatch in the classes to which the query features and the support features belong during the aggregation process, as shown in Fig. 1(b). Since the label of the aggregated features is only determined by the query features, the classifier will be forced to ignore the support features and cannot recognize novel classes correctly after model fine-tuning.

3) Target Category Confusion: SAR object detection usually focuses more on model-level fine-grained recognition tasks. For subtargets of the same class, their SAR images vary with the observation angle, leading to large intraclass differences. In addition, due to the scarcity of training samples for novel classes in few-shot SAR object detection, the classifier at the backend of the model cannot be fully trained, resulting in confusion between different novel classes, as well as between novel classes and their corresponding similar base classes, as shown in Fig. 1(c).

To address these challenges, we propose a Few-Shot SAR object Detection model with Dynamic perception region proposal network (RPN) and category knowledge Decoder, namely FSD^3, which comprises a dynamic perception RPN (DP-RPN), a class knowledge decoder (CKD), and a dual-constrained detection head (DCDH). Specifically, to tackle the challenge of foreground–background confusion, the DP-RPN first calculates the region-aware mask based on the support feature maps and the query feature maps and generates dynamic convolution kernels for the corresponding classes using support feature maps. Subsequently, it applies dynamic region convolution [42] to the query feature maps by the dynamic convolution kernels and the region-aware mask, thereby infusing region features of the query feature maps with class semantics. By this means, we alleviate the bias toward base classes and mitigate the impact of background clutter and speckle noise on region proposal generation. To resolve the mismatch between features obtained by aggregation networks and their corresponding labels, the CKD first feeds the support features of each class into the attention module [43], [44] for feature encoding and then decodes the encoded class knowledge (i.e., support features of different classes) with the query RoI features as prompts, thereby maintaining the consistency between features of support samples and their corresponding labels. To mitigate class confusion induced by high interclass similarity and large intraclass variance, the DCDH employs large margin cosine loss [45] to increase the distance between different classes in the high-dimensional feature embedding space, thereby achieving interclass separation. Simultaneously, it utilizes center calibration loss (CCL) to reduce the distance between projections of the same class in the high-dimensional feature space, thereby achieving intraclass compression. Eventually, the classification performance under the condition of few-shot detection is improved.

The main contributions of this article can be summarized as the following three aspects.

1) The DP-RPN is proposed to strategically integrate semantic information from support features into diverse spatial locations of the query feature maps, thereby significantly mitigating background clutter interference in the SAR imagery and achieving effective foreground–background separation.

2) The CKD is proposed to directly decode the encoded category knowledge by utilizing the query RoI features as prompts, which enables the query features to play a dominant role during feature aggregation and mitigates the mismatch between aggregated features and their corresponding labels, thereby maintaining the consistency between features of support samples and their corresponding labels.

3) The DCDH is proposed to enhance the separation between different classes via the large margin cosine loss [45] and reduce the intraclass distances via the CCL, thereby achieving interclass separation and intraclass compactness, and significantly improving the classification accuracy.

The rest of this article is organized as follows. Section II introduces related work. Section III presents basic concepts of few-shot object detection and introduces the proposed FSD^3 model. Section IV provides the experimental results and analysis. Section V concludes this article.

II. Related Work

A. SAR Object Detection

SAR target detection methods can be categorized into two primary paradigms: traditional methods and deep learning-based methods.

1) Traditional Methods: Traditional SAR object detection and recognition primarily follow the three-stage processing

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pipeline developed by MIT Lincoln Laboratory. This pipeline consists of object detection, object discrimination, and object recognition. Conventional detection approaches are predominantly based on clutter statistical analysis and threshold estimation via constant false alarm rate (CFAR) algorithms. In the design of CFAR detectors, various methods such as CA-CFAR [46] and OS-CFAR [47] have been developed to address relatively straightforward scenarios. To address complex scenarios, researchers have introduced advanced CFAR techniques, including ICS-CFAR [48] and Super-pixel CFAR [49]. Regarding clutter statistical modeling, the G0, Weibull, and K distributions are widely adopted for homogeneous clutter backgrounds [50], [51], [52].

2) Deep learning-Based Methods: They accomplish SAR object detection in an end-to-end manner by constructing deep networks, thereby avoiding the cumbersome traditional three-level processing pipeline and becoming the mainstream of SAR object detection. These approaches are categorized into CNN-based methods [3], [10], [14], [18], [53], [54], [55] and Transformer-based methods [11], [12], [15], [16], [17], [56], [57]. The former leverages anchor boxes or anchor points for label assignment and relies on the strong local feature extraction capability of CNNs to achieve efficient and precise object detection. For instance, the feature enhancement pyramid [54] is designed to amplify the weak feature responses of small-size ship targets in SAR imagery, the DenoDet [10] is crafted to mitigate the interference of speckle noise in SAR images, and multistage filtering-enhanced pretraining [3] is devised to address the disparity between SAR and optical imagery. The latter, evolving from the DETR series [16], [17], [56], transforms SAR object detection into a set prediction problem via the Hungarian algorithm, thereby eschewing post-processing steps such as NMS and realizing end-to-end SAR object detection. Current methods primarily focus on resolving the confusion between SAR targets and background noise, as well as speckle noise, which design multilevel denoising techniques at the input layer, the feature layer, the query layer [11], and so on. Moreover, contrastive denoising is proposed based on noisy query reconstruction [12].

B. Few-Shot Object Detection

Few-shot object detection tasks aim at achieving high-precision object detection with limited training data, and current methodologies mainly include two frameworks: metalearning and transfer learning.

1) Metalearning-Based Few-Shot Object Detection: This paradigm initially trains a metalearner containing metaknowledge on a training set comprising base class data and then fine-tunes on novel class data to acquire the detection capability for novel classes. Meta-YOLO [39] processes support images through a reweighting network, which adopts shared weights as the feature extractor and generates channelwise reweighting vectors to modulate the query feature maps. Later, Meta region-based convolutional neural network (RCNN) [34] adapts this method to two-stage detectors, leveraging its architectural advantages for enhanced precision. In contrast, FsDetView [35] abandons channelwise multiplicative aggregation and guides the network to learn optimal feature fusion mappings automatically. The multirelation detection architecture in Attention RPN [36] integrates metric learning, thereby enabling category discrimination through comprehensive analysis of local, global, and patch relations between query and support features. To meet the challenge of data scarcity, VFA [37] employs VAE [58] to reconstruct support features in the latent space, reducing dependence on limited support samples and improving novel-class generalization. For few-shot SAR object detection, Zhou et al. [40] designed a feature extractor for SAR images and adopted an encoder to complete the interaction between support and query features in the feature subspace. To address the instability of class centers in few-shot scenarios, Zhou et al. [41] introduce dense metafeature aggregation to stabilize sample distributions and improve detection robustness.

2) Transfer Learning-Based Few-Shot Object Detection: This paradigm preserves the detector architecture while enabling novel class adaptation through two-stage fine-tuning, treating the base class data as the source domain and the novel class data as the target domain. Notably, TFA [27] achieves competitive novel class detection performance via parameter freezing of the backbone and sequential fine-tuning of the classification and regression heads. To address the challenges of single-scale representation for novel classes, MPSR [28] introduces an auxiliary branch for multiscale object generation, effectively augmenting the samples and facilitating FPN [59] training. Sun et al. [29] proposed a robust feature representation for few-shot object detection by combining supervised contrastive learning. Furthermore, Xu et al. [60] tackle the challenge of insufficient overlap between region proposals and ground truth in few-shot scenarios by leveraging VAEs to reconstruct the RoI feature matrices of region proposals, which effectively enhances the IoU values and optimizes classifier training. In addition, Ren et al. [61] proposed the Confucius trilearning paradigm, which employs two discriminators and one generator to learn from good and bad samples, significantly reducing the reliance of the model on labels. To address the sample imbalance between base and novel classes, Wu et al. [62] adopted an adversarial feature training strategy that generates adversarial samples to calibrate the decision boundary between base and novel classes, thereby mitigating category confusion.

Although metalearning has demonstrated superior performance and more flexibility to transfer learning in few-shot object detection, existing approaches have overlooked the inherent challenges posed by speckle noise and background clutter in the SAR imagery, thereby making the model prone to neglecting novel classes. To address this, we introduce DP-RPN to enhance the recall rate of novel classes. Second, existing approaches usually suffer from category label mismatch during feature aggregation, as illustrated in Fig. 1(b). To mitigate this, we propose CKD, which utilizes query features as prompts to decode the category-specific knowledge from support prototypes, thereby ensuring the consistency between features and their assigned labels. Finally, considering the critical yet overlooked challenge of fine-grained model-level discrimination in SAR object detection, we develop DCDH to enhance the discriminability among classes.

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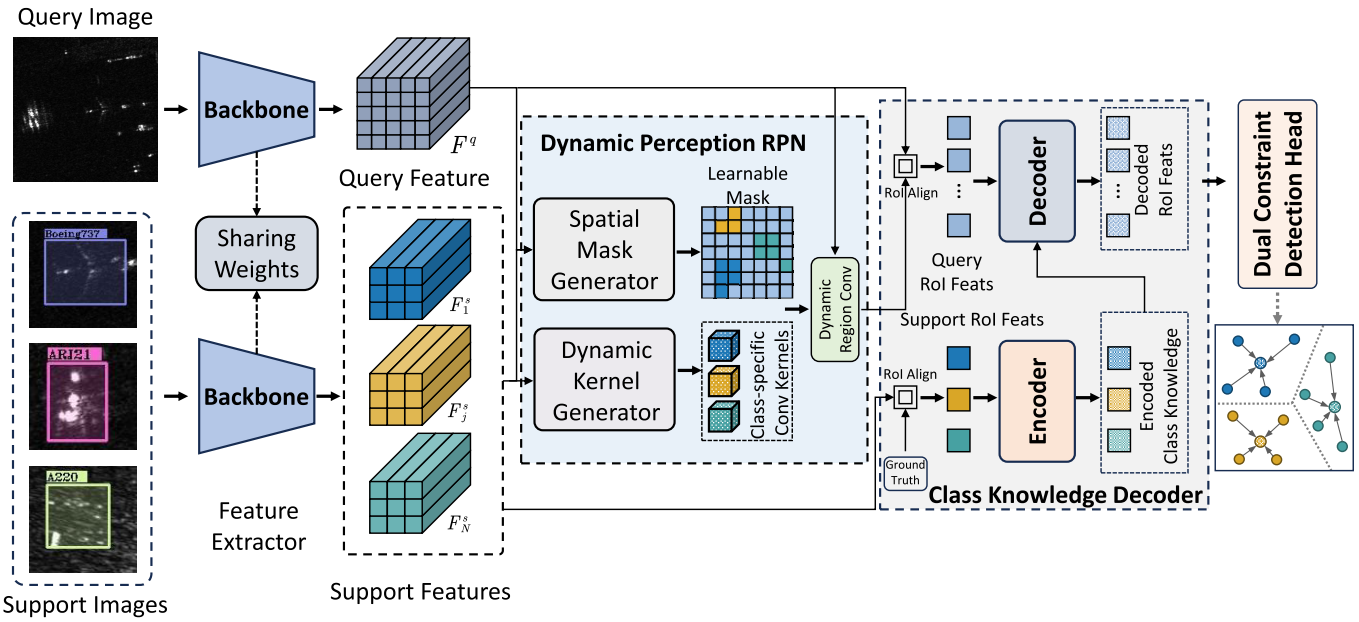


Fig. 2. Overall architecture of the proposed model. Initially, the backbone is employed to extract feature maps from both the query and support images, which are then fed into the DP-RPN to generate region proposals for the query image. After RoI Align [63], the feature matrix is further transformed into RoI feature vectors. Finally, the CKD utilizes the query RoI feature vectors as prompts to decode class knowledge, and the decoded features are then fed into the DCDH to yield the final detection results.

III. Method

A. Problem Definition

In few-shot object detection, the complete class space $\mathcal{C}$ is partitioned into disjoint subsets: base classes $\mathcal{C}_{\text{base}}$ and novel classes $\mathcal{C}_{\text{novel}}$, satisfying $\mathcal{C}_{\text{base}} \cup \mathcal{C}_{\text{novel}} = \mathcal{C}$ and $\mathcal{C}_{\text{base}} \cap \mathcal{C}_{\text{novel}} = \varnothing$. On this basis, the training data can be further categorized into the base training set $\mathcal{D}_{\text{base}}$ and the novel training set $\mathcal{D}_{\text{novel}}$. The base training set $\mathcal{D}_{\text{base}}$ encompasses a substantial quantity of labeled instances of the base classes $\{\boldsymbol{x}_i, \boldsymbol{y}_i\}$, where $\boldsymbol{x}_i$ represents the ith training image and $\boldsymbol{y}_i = \{(c_j^i, \boldsymbol{b}_j^i)\}$ represents the labels and bounding boxes of the target instances within the ith training image. By K-shot few-shot object detection, we mean that each novel class within the novel training set $\mathcal{D}_{\text{novel}}$ includes K annotated instances distributed across $\hat{K}$ $(1 \leq \hat{K} \leq K)$ images.

Given the base training set $\mathcal{D}_{\text{base}}$ and the novel training set $\mathcal{D}_{\text{novel}}$, the K-shot few-shot object detection task is accomplished through a two-stage training process in the metalearning-based paradigm. In the first stage, the model undergoes extensive training on the richly annotated base training set $\mathcal{D}_{\text{base}}$ to establish a robust base detection model. In the second stage, with partial weights frozen in the base detection model, the network is fine-tuned using a balanced training set $\mathcal{D}_{\text{balance}}$ to obtain the final model. To generate $\mathcal{D}_{\text{balance}}$, we sample K training instances for each class from $\mathcal{D}_{\text{base}}$ to form $\mathcal{D}_{\text{base}}^-$, which is then combined with $\mathcal{D}_{\text{novel}}$, i.e., $\mathcal{D}_{\text{balance}} = \mathcal{D}_{\text{novel}} \cup \mathcal{D}_{\text{base}}^-$.

B. Overall Structure

The overall architecture of FSD^3 is illustrated in Fig. 2, which mainly includes three modules: the DP-RPN, the CKD, and the DCDH. Since SAR images typically encompass large scenes with relatively small targets, directly utilizing the unprocessed support images would introduce a significant amount of irrelevant background information. In view of this, the support images are cropped from the original image and resized to a dimension of $L \times L$. Subsequently, the query image X^q and the support images X_n^s are fed into the weight-shared backbone $\mathcal{B}_\Theta(\cdot)$ for feature extraction, where $n \in (1, N)$ denotes the index of classes. After that, the query feature maps F^q and support feature maps F_n^s are obtained

$$F^q = \mathcal{B}_\Theta(X^q), \qquad F_n^s = \mathcal{B}_\Theta(X_n^s) \tag{1}$$

where Θ denotes parameters of the backbone.

Subsequently, the query feature maps and support feature maps are fed into the proposed DP-RPN. By infusing the semantic information of different classes from the support feature maps into various spatial locations of the query feature maps, the issue of low recall rates for novel classes due to insufficient samples is alleviated. For the region proposals output by the DP-RPN, RoI Align [63] is employed to obtain their corresponding RoI feature matrices, which are then subjected to further feature extraction to derive the RoI feature vectors for the query image and support images, denoted as f_m^q and f_n^s, respectively, where $m = 1, 2, \ldots, M$ with M representing the number of region proposals in the query image.

Next, the CKD utilizes the query RoI feature vector f_m^q as a prompt to decode the encoded category knowledge and generate the decoded query RoI feature vector f_m^d. Finally, f_m^d is fed into the DCDH to generate the detection results.

The structural details and processing flow of the aforementioned modules will be elaborated on in the following.

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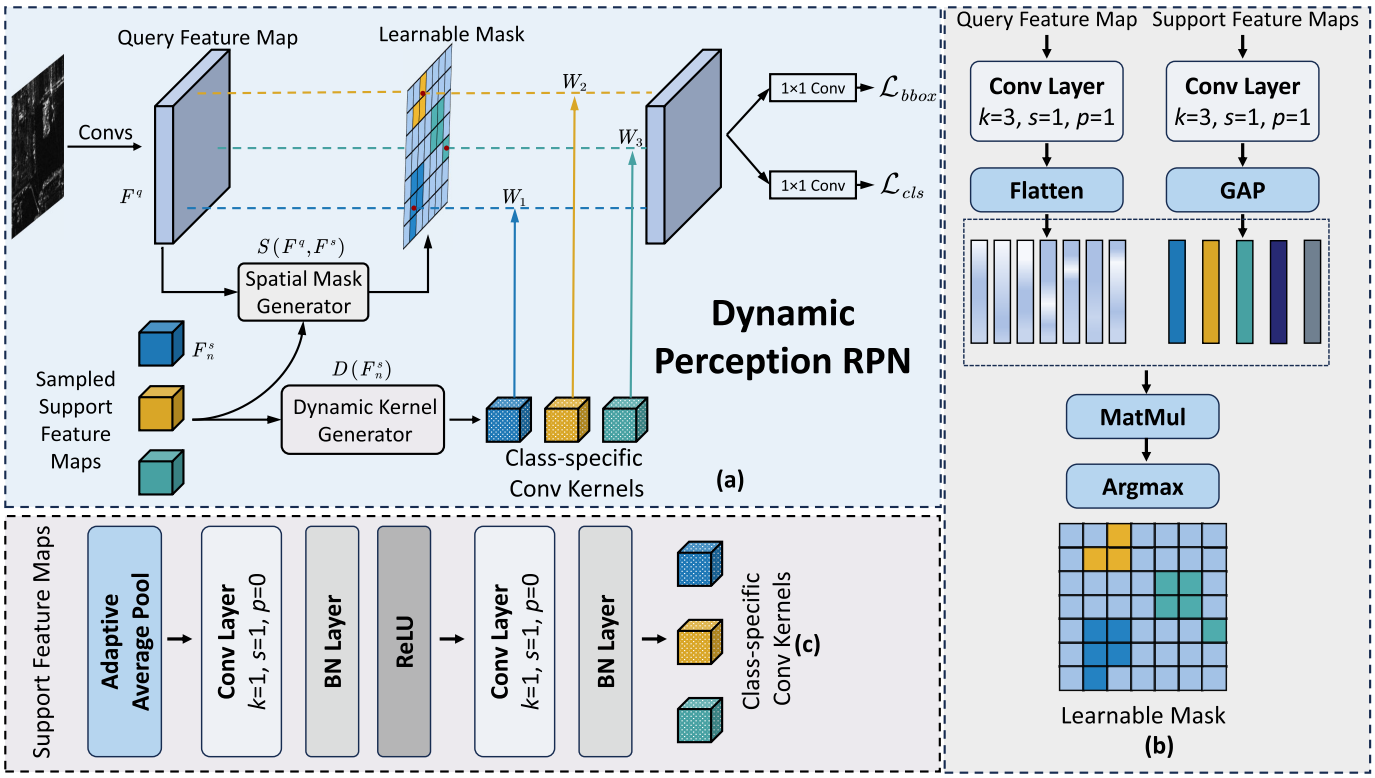


Fig. 3. Schematic of the proposed DP-RPN. (a) Overall architecture of DP-RPN. (b) Network structure of the spatial mask generator. (c) Network structure of the dynamic kernel generator.

C. Dynamic Perception RPN

During base training, the RPN [14] is trained with base-class targets as foreground. Consequently, a bias toward the base-class targets is introduced, which makes RPN prioritize base-class targets and overlook novel-class targets during region proposal generation. In addition, the background clutter for SAR images further hinders high-quality region proposals for novel classes. To address this, the DP-RPN modulates the feature maps by incorporating class semantic information into the query feature maps, thereby enabling the generation of high-quality region proposals for novel-class targets without fine-tuning the parameters of RPN.

The architecture of the proposed DP-RPN is illustrated in Fig. 3, which enhances the conventional RPN by incorporating a spatial mask generator and a dynamic kernel generator. As depicted in Fig. 3(b), given the query feature map F^q and the support feature maps F_n^s, the spatial mask generator produces a learnable region-aware mask with the same size as the query feature map. Specifically, it flattens the query feature map along the spatial dimensions to obtain $\tilde{F}^q$ and then applies global average pooling to the support feature map to derive the global features of support images $\tilde{F}_n^s$, i.e.,

$$\tilde{F}^q = \text{Flatten}\big(\text{Conv}_{3\times3}\big(F^q\big)\big) \tag{2}$$

$$\tilde{F}_n^s = \text{GAP}\big(\text{Conv}_{3\times3}\big(F_n^s\big)\big) \tag{3}$$

where Flatten$(\cdot)$ denotes flattening along the spatial dimension, GAP$(\cdot)$ represents global average pooling, and $\text{Conv}_{3\times3}(\cdot)$ signifies a 3×3 convolution operation.

Subsequently, the inner product of $\tilde{F}^q$ and $\tilde{F}_n^s$ is computed successively to determine the similarity between local features of $\tilde{F}^q$ and the support features $\tilde{F}_n^s$ of class n, which is then reshaped to obtain the similarity map S_n

$$S_n = \text{reshape}\Big(\big(\tilde{F}^q\big)^T \times \tilde{F}_n^s\Big) \tag{4}$$

which shares the same size as the query feature map.

Finally, the entry of the region-aware mask Ω at position (h, w) is given by

$$\Omega(h, w) = \text{argmax}(S_1(h, w), S_2(h, w), \ldots, S_N(h, w)) \tag{5}$$

where argmax$(\cdot)$ denotes taking the index of the maximum value.

Meanwhile, the dynamic kernel generator processes the support feature map F_n^s into a class-specific dynamic convolution kernel W_n through the dynamic kernel generation network illustrated in Fig. 3(c). Specifically, it applies average pooling to the support feature map F_n^s, followed by a 1×1 convolution to reduce the dimension to $(1/16)$ of the original size. Then, another 1×1 convolution is employed to increase the dimensionality to $c_{\text{in}} \times c_{\text{out}} \times k^2$ and obtain W_n, where c_{in} and c_{out} represent the input and output channels of the kernel, respectively, and k denotes the kernel size.

Finally, for regions defined by Ω in the query feature map F^q, the DP-RPN performs dynamic region convolution [42] with convolution kernel W_n as follows:

$$Y(h, w, o) = \sum_{c=1}^{c_{in}} \sum_{i=\lceil -\frac{k}{2} \rceil}^{\lfloor \frac{k}{2} \rfloor} \sum_{j=\lceil -\frac{k}{2} \rceil}^{\lfloor \frac{k}{2} \rfloor} V(h, w, o, i, j, c) \tag{6}$$

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$$V(h,w,o,i,j,c) = F^q(h+i, w+j, c) \cdot W^{(o)}_{\Omega(h,w)}(i,j,c) \tag{7}$$

where $Y(h,w,o)$ denotes the feature value of the output feature map at (h,w) for oth channel, while $F^q(h,w,c)$ denotes the feature value of the query feature map at (h,w) for the cth channel.

To enhance the generalization capability of the DP-RPN, we train it by randomly sampling the support features. Specifically, we randomly select $\hat{N}$ classes from the N support feature classes to form a set $\hat{\mathcal{C}}$, where $1 \le \hat{N} \le N$. Then, targets within $\hat{\mathcal{C}}$ are treated as foreground, while those outside $\hat{\mathcal{C}}$ are regarded as background.

After transferring the global semantics of support features to local features of the query feature map, we feed them to two parallel convolutional layers to compute the classification loss and the bounding box regression loss, respectively. The output region proposals are then fed into the RoI Head for further class refinement and bounding box adjustment.

D. Class Knowledge Decoder

In metalearning-based few-shot object detection, it is essential to aggregate the support and query features effectively. Existing feature aggregation methods sample support features from a single class and then perform dot product, subtraction, concatenation, and so on with the query features [34], [35]. In the training stage, however, class labels assigned to the feature vectors are solely determined by the query features. The oversight of support samples can lead to a mismatch between the aggregated feature vectors and their assigned labels, thereby confusing the classifier, as illustrated in Fig. 1(b). Furthermore, these methods randomly sample support features from a single class during training but sequentially sample support features from all classes during inference. Such inconsistency between the training and inference pipelines misaligns the training objective function and the desired detection outcomes, thereby compromising the model's stability.

To address this, we propose CKD. Rather than explicitly sampling support features from specific classes, it utilizes query feature vectors as prompts to adaptively select and aggregate support features. As illustrated in Fig. 4, after RoI align [63], we obtain query RoI feature matrices F^q_{roi} and support RoI feature matrices F^s_{roi}. Due to the limited long-range modeling capability of convolutional operations, distant targets in the SAR image often lack sufficient feature interaction. In view of this, we perform instance-level feature interaction on RoI features to enhance their representational power. Specifically, multihead attention is applied to F^q_{roi} and F^s_{roi}, respectively, to facilitate semantic interaction between the RoI features; then, the semantic information is reassigned to the corresponding RoI feature matrices

$$\tilde{F}^q_{\text{roi}} = \text{FFN}\left(F^q_{\text{roi}}\right) \times \text{MultiAtten}\left(\text{GAP}\left(F^q_{roi}\right)\right) \tag{8}$$

$$\tilde{F}^s_{\text{roi}} = \text{FFN}\left(F^s_{\text{roi}}\right) \times \text{MultiAtten}\left(\text{GAP}\left(F^s_{\text{roi}}\right)\right) \tag{9}$$

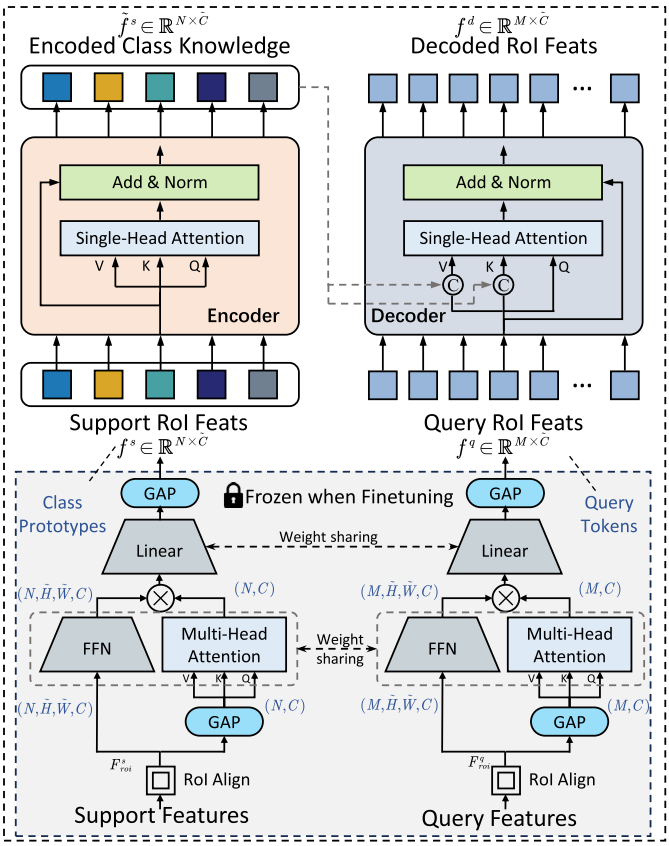


Fig. 4. Category knowledge decoder. By utilizing the query RoI features as prompts, it adaptively selects and aggregates the support features, effectively addressing the mismatch between features and labels while ensuring consistency between the training and inference phases of the model.

where FFN$(\cdot)$ denotes the feedforward network and MultiAtten$(\cdot)$ denotes the multihead self-attention mechanism. The weights involved in (8) and (9) are shared to ensure that the targets in the support and query images are mapped into the same feature space.

Given $\tilde{F}^q_{\text{roi}}$ and $\tilde{F}^s_{\text{roi}}$, a 1×1 convolutional linear mapping followed by global average pooling is applied to derive the query RoI feature vector (i.e., the query token f^q) and the support RoI feature vector (i.e., the class prototype f^s).

Subsequently, f^s are encoded into category knowledge $\tilde{f}^s$ through single-head self-attention

$$f^s_q = f^s W^Q_1,\ f^s_k = f^s W^K_1,\ f^s_v = f^s W^V_1 \tag{10}$$

$$\tilde{f}^s = \text{LayerNorm}\left(\text{softmax}\left(\frac{f^s_q \times \left(f^s_k\right)^T}{\sqrt{\tilde{C}}}\right) f^s_v + f^s\right) \tag{11}$$

where LayerNorm$(\cdot)$ denotes the layer normalization [64].

It is worth noting that if only cross-attention is applied for the query and support RoI feature aggregation, the detection head would only accept linear combinations of the support features, thereby diminishing the contribution of query RoI features and degrading the detection performance. To address this issue, we explicitly incorporate the query RoI feature vectors into the affinity matrix during feature decoding.

Specifically, the query RoI feature vector is projected into the corresponding Q, K, and V spaces to obtain f^q_q, f^q_k, and

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f_v^q, respectively,

$$f_q^q = f^q W_2^Q, \quad f_k^q = f^q W_2^K, \quad f_v^q = f^q W_2^V. \tag{12}$$

Simultaneously, the category knowledge $\tilde{f}^s$ is projected into the K and V spaces, yielding $\tilde{f}_k^s$ and $\tilde{f}_v^s$

$$\tilde{f}_k^s = \tilde{f}^s W_2^K, \quad \tilde{f}_v^s = \tilde{f}^s W_2^V. \tag{13}$$

Then, we compute the affinity matrix A_m by incorporating the Q space projection f_{qm}^q and the K space projection f_{km}^q of mth RoI feature vector, along with the K space projections $\tilde{f}_k^s$ of all the encoded category knowledge, i.e.,

$$A_m = \text{softmax}\left(\frac{\text{Concat}\left(f_{qm}^q \times \left(\tilde{f}_k^s\right)^T, f_{qm}^q \times \left(f_{km}^q\right)^T\right)}{\sqrt{C}}\right). \tag{14}$$

In order to combine features in the V space projections based on the weights in the affinity matrix A_m, the affinity matrix A_m should be partitioned since both the mth query RoI feature vector and all the encoded category knowledge are projected into the V space for feature composition; specifically,

$$A_m^s, A_m^q = \text{Split}(A_m) \text{ with size } [N, 1]. \tag{15}$$

Finally, feature composition is performed by applying A_m^s and A_m^q to weight and combine the V space projections of both the query RoI features and encoded category knowledge, i.e.,

$$f_m^d = \text{LayerNorm}\left(A_m^s \times \tilde{f}_v^s + A_m^q \times f_{vm}^q + f_m^q\right). \tag{16}$$

After that, f^d is fed into the detection head to obtain the detection results.

E. Dual-Constrained Detection Head

To preserve the detection performance of base classes, most few-shot object detectors freeze parameters of the backbone during fine-tuning, thereby hindering the effective feature extraction of novel classes. Given the significant intraclass variance and high interclass similarity in SAR images, the training of a highly discriminative feature extraction network is significant yet challenging. To address this issue, the DCDH is developed to explicitly constrain the feature representation of different classes in the feature space by constructing two losses: the large-margin cosine loss is employed to increase the distance between different classes, and the CCL is utilized to reduce the distance between query and support features for the same class.

When bias is ignored, the decision boundary of the softmax classifier in angular space can be expressed as follows:

$$||w_i|| \cos\theta_i = ||w_j|| \cos\theta_j \tag{17}$$

where w_i and w_j denote the classifier weights for class i and j, respectively, and θ_i and θ_j represent the angles between the feature vector, w_j, and w_j. According to (17), the decision boundary of the softmax classifier depends not only on the angles between the weight and input features but also on the norms of the classifier weights. Consequently, the decision boundary varies with the magnitude of the weight norms. Under few-shot conditions, the norms of the novel-class classifier weights struggle to maintain balance with those of the base-class classifiers. This imbalance leads to negative margins in the decision boundaries across different classes. Existing literature [27], [29] employed the cosine classifier to eliminate the influence of classifier weight norms on decision boundaries, thereby enforcing zero margins between different classes and achieving performance improvements. However, SAR image targets exhibit high interclass similarity. The zero-margin decision boundaries obtained solely through the cosine classifier remain insufficient for accurate classification of SAR targets under the few-shot condition.

To address this, the large margin cosine classifier [45] is employed to classify the decoded feature f^d, with the loss $\mathcal{L}_{\text{lmc}}$ expressed as

$$\mathcal{L}_{\text{lmc}} = \frac{1}{M}\sum_m -\log\frac{e^{s(\cos(\theta_{y_m},m)-\phi)}}{e^{s(\cos(\theta_{y_m},m)-\phi)} + \sum_{j\neq y_m} e^{s\cos(\theta_j,m)}} \tag{18}$$

where

$$\cos\left(\theta_j, m\right) = \tilde{w}_j^T \tilde{f}_m^d \tag{19}$$

$$\tilde{f}_m^d = \frac{f_m^d}{||f_m^d||} \tag{20}$$

$$\tilde{w}_j = \frac{w_j}{||w_j||} \tag{21}$$

where w_j is the classifier weight corresponding to class j, ϕ is the predefined margin parameter, and s is the predefined scaling factor, which is set to 30.

During inference, the classifier assigns the mth feature vector to class n if the following condition is satisfied:

$$C_n : \cos(\theta_n, m) \geq \max_{j\in\{1,\ldots,N\}, j\neq n} \cos\left(\theta_j, m\right) + \phi. \tag{22}$$

To address the large intraclass variance, the CCL is introduced to explicitly reduce the distance between query features and their class prototypes. This loss is applied to the support and query features before category knowledge decoding and can be expressed as

$$\mathcal{L}_{\text{ccl}} = \frac{1}{M}\sum_{m=1}^{M}\left(e^{||f_m^q - f_{y_m}^s||} - 1\right) \tag{23}$$

where $f_{y_m}^s$ represents the prototype of class y_m obtained from the support samples.

By integrating the large margin cosine loss and the CCL, we can enhance interclass differences and reduce intraclass variance through dual-feature constraints, thereby improving the performance and robustness of the classifier.

F. Model Training

The training of FSD^3 comprises two stages, i.e., base training and few-shot fine-tuning, and the loss function is defined as

$$\mathcal{L} = \mathcal{L}_{\text{proposal}} + \beta\mathcal{L}_{\text{lmc}} + \mathcal{L}_{\text{bbox}} + \mathcal{L}_{\text{meta}} + \gamma\mathcal{L}_{\text{ccl}} \tag{24}$$

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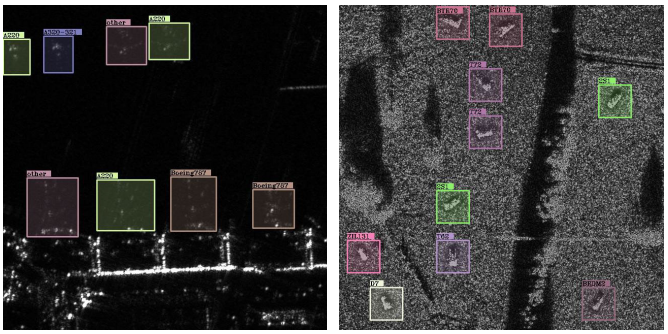


Fig. 5. Illustrative examples of images from the SAR-Aircraft-1.0 and MSTAR-Fewshot datasets.

TABLE I
CLASS SPLITS FOR THE TWO DATASETS

Dataset	Split	Base Classes	Novel Classes
SAR-AIRCraft-1.0	1	A220, A320, A330 B787, Other	ARJ21, B737
	2	A220, A320, A330, B737, Other	ARJ21, B787
	3	A320, A330, B737, B787, Other	A220, ARJ21
MSTAR-Fewshot	1	2S1, T62, ZIL131, D7, ZSU234, BRDM2, BTR60	BMP2, BTR70, T72
	2	BMP2, BTR70, T72, D7, ZSU234, BRDM2, BTR60	2S1, T62, ZIL131
	3	BMP2, BTR70, T72, 2S1, T62, ZIL131, BTR60	D7, ZSU234, BRDM2

where $\mathcal{L}_{\text{proposal}}$ represents the region proposal loss for DP-RPN and takes the similar form as that adopted in conventional RPN [14]; $\mathcal{L}_{\text{lmc}}$ represents the large margin cosine loss [45], with the weight factor β set to 1 in this article; $\mathcal{L}_{\text{bbox}}$ represents the bounding box regression loss; $\mathcal{L}_{\text{meta}}$ represents the metaloss employed during Meta RCNN [34] training; and $\mathcal{L}_{\text{ccl}}$ represents the CCL, with the weight factor γ set to 10.

The training of FSD^3 includes two stages: base training and few-shot fine-tuning. The former utilizes a substantial number of base-class samples to train the base model, while the latter further fine-tunes the model using K training samples, i.e., K shots for each class. During fine-tuning, all parameters of the backbone, DP-RPN, and partial parameters of the CKD are frozen to preserve the detection capability of base classes. Furthermore, during the training stage of FSD^3, multiple query images are sampled per iteration, whereas only a single support image per class is selected for model training to alleviate computational demands.

IV. EXPERIMENTS

In this section, we first introduce the two datasets utilized in the experiments, namely, SAR-Aircraft-1.0 [6] and MSTAR-Fewshot. Subsequently, we present standard evaluation metrics commonly adopted in few-shot object detection tasks. Finally, we conduct comparative performance analysis between the proposed model and several state-of-the-art (SOTA) few-shot object detection approaches and demonstrate the efficacy of the proposed modules through ablation studies.

A. Dataset Description

1) SAR-Aircraft-1.0: SAR-AIRcraft-1.0 [6] is a fine-grained detection dataset designed for SAR aircraft detection in large-scale complex scenarios. The data are collected by the Gaofen-3 satellite, employing spotlight imaging mode with HH polarization. The imaging scenes encompass major civilian airports such as Shanghai Hongqiao Airport, Beijing Capital Airport, and Taiwan Taoyuan Airport, with a resolution of 1 × 1 m. Through cropping and postprocessing, the dataset comprises 4368 images and 16 463 aircraft instances. These images encompass four distinct sizes: 800 × 800, 1000 × 1000, 1200 × 1200, and 1500 × 1500. The aircraft are categorized into seven specific types: A220, A320/321, A330, ARJ21, Boeing 737, Boeing 787, and Other. A selection of sample images from the dataset is illustrated in Fig. 5.

To validate the generalization capabilities of the proposed model, three distinct configurations for novel and base class splits are designed and outlined in Table I. For each novel and base class split, five few-shot scenarios are established, including 1-, 3-, 5-, 10-, and 30-shot settings. For any K-shot scenario, ten random seeds are utilized to sample training data during model fine-tuning. Ultimately, results from a single run and average results across the ten random seeds are presented.

2) MSTAR-Fewshot: Since the available SAR object detection datasets have a limited number of classes, there may be only one novel class in a task after class division, thereby introducing uncertainty and instability during performance evaluation. To tackle this, the MSTAR-Fewshot dataset is synthesized by integrating the MSTAR dataset [65] with large-scene SAR images as the background. The MSTAR dataset, curated and released by Sandia National Laboratories in the United States, consists of ten types of military vehicles. For the MSTAR-Fewshot dataset, it comprises 900 SAR images with a size of 640 × 640, including 540 training images and 360 test images. A selection of sample images is depicted in Fig. 5.

Similar to SAR-Aircraft-1.0, we establish three distinct class splits for the MSTAR-Fewshot dataset, as outlined in Table I. For each class split, five few-shot scenarios are configured: 1-shot, 2-shot, 3-shot, 5-shot, and 10-shot. To ensure stable experimental results, for each K-shot scenario, ten different random seeds are sampled to generate the training data for model fine-tuning. Similarly, results from a single run and average results across the ten random seeds are presented.

B. Implementation Details

For the SAR-Aircraft-1.0 [6] dataset, all images are uniformly resized to 1024 × 1024 using coordinates mapping and bilinear interpolation [66]. During base training, the batch size is set to 8, the learning rate is initialized as 0.005, and the SGD optimizer is employed for 24 000 iterations. The learning rate is reduced to 0.1 of its original value at the 12 000th and 22 000th iterations, respectively. During few-shot fine-tuning,

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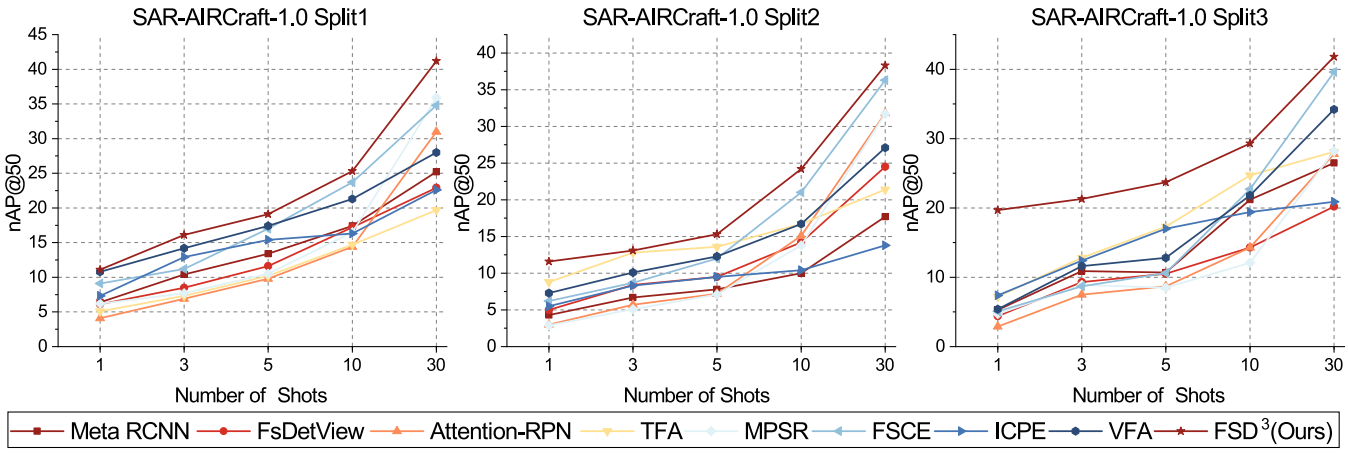


Fig. 6. Experimental results on the SAR-AIRCraft-1.0 dataset under multiple runs with different random seeds. It can be observed that the proposed method achieves the best novel class detection performance across all experimental settings.

the batch size remains as 8, while the learning rate is adjusted to 0.0005. For the SGD optimizer, the model is fine-tuned for 2500, 3500, 6000, 10 000, and 12 000 iterations under the 1-, 3-, 5-, 10-, and 30-shot settings, respectively.

Similarly, for the MSTAR-Fewshot dataset, all images are uniformly resized to 640 × 640 using coordinate mapping and bilinear interpolation. During base training, the batch size is set to 8, the learning rate is initialized at 0.005, and the SGD optimizer is employed for 8000 iterations. The learning rate was reduced to 0.1 of its original value at the 6000th iteration. During few-shot fine-tuning, the batch size remains as 8, while the learning rate is adjusted to 0.001. For the SGD optimizer, the model is fine-tuned for 2000, 3000, 3500, 4000, and 5000 iterations under the 1-, 2-, 3-, 5-, and 10-shot settings, respectively.

C. Evaluation Metric

In this article, we evaluate the model's performance by calculating the mean average precision (mAP) across various thresholds. Specifically, different confidence thresholds τ are set to obtain the precision $P(\tau)$ and recall $R(\tau)$, and multiple pairs of $P(\tau)$ and $R(\tau)$ are applied to calculate the precision–recall (P–R) curve $P(R)$. Consequently, the mAP is defined as

$$\text{AP} = \int_0^1 P(R)\text{dR} \tag{25}$$

$$\text{mAP} = \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}} \text{AP}_c \tag{26}$$

where AP_c denotes the average precision (AP), c denotes the class index, and $\mathcal{C}$ denotes the set of all classes.

To evaluate the detection performance of novel and base classes separately, the novel AP (nAP) and base AP (bAP) are further computed

$$\text{nAP} = \frac{1}{|\mathcal{C}_{\text{novel}}|}\sum_{c\in\mathcal{C}_{\text{novel}}} \text{AP}_c \tag{27}$$

$$\text{bAP} = \frac{1}{|\mathcal{C}_{\text{base}}|}\sum_{c\in\mathcal{C}_{\text{base}}} \text{AP}_c \tag{28}$$

where $\mathcal{C}_{\text{novel}}$ and $\mathcal{C}_{\text{base}}$ denote the sets of novel and base classes.

In addition to mAP, we introduced average recall (AR) as an additional metric during the ablation study of DP-RPN to evaluate the recall capability of different region proposal generation methods. The AR computation is formulated as follows:

$$\text{AR} = 2\int_{0.5}^{1} R(u)\text{du} \tag{29}$$

where u denotes the IoU threshold.

D. Experimental Results

In this part, performance of the proposed FSD^3 is compared with SOTA few-shot object detection methods, including metalearning-based methods, i.e., Meta RCNN [34], FsDetView [35], Attention-RPN [36], VFA [37], and ICPE [38], and transfer learning-based methods, i.e., TFA [27], MPSR [28], and FSCE [29]. Since the performance in novel class detection relies heavily on the choice of training samples during fine-tuning, and SAR targets exhibit significant intra-class variation, the detection performance for novel classes fluctuates considerably. To address this, a set of training samples is first generated to form a fixed set for fine-tuning, and then, the single-run results are obtained and presented in the upper parts of Tables II and III. Subsequently, by setting different random seeds, ten distinct sample sets are generated for fine-tuning to obtain the average detection results, as shown in the lower parts of Tables II and III. By this means, a comprehensive evaluation of the performance can be achieved. Figs. 6 and 7 illustrate the variation of novel class detection performance with the number of shots on the SAR-Aircraft-1.0 and MSTAR-Fewshot datasets, respectively.

1) Result of SAR-Aircraft-1.0: Table II presents the experimental results on the SAR-AIRcraft-1.0 dataset, where red indicates the optimal performance and blue indicates the suboptimal performance. From the single-run results, it is observed that the proposed method achieves the best performance under most experimental settings, only slightly underperforming TFA [27] in the 3-shot condition under class split 2 (−1.3%) and FSCE [29] in the 10-shot condition under

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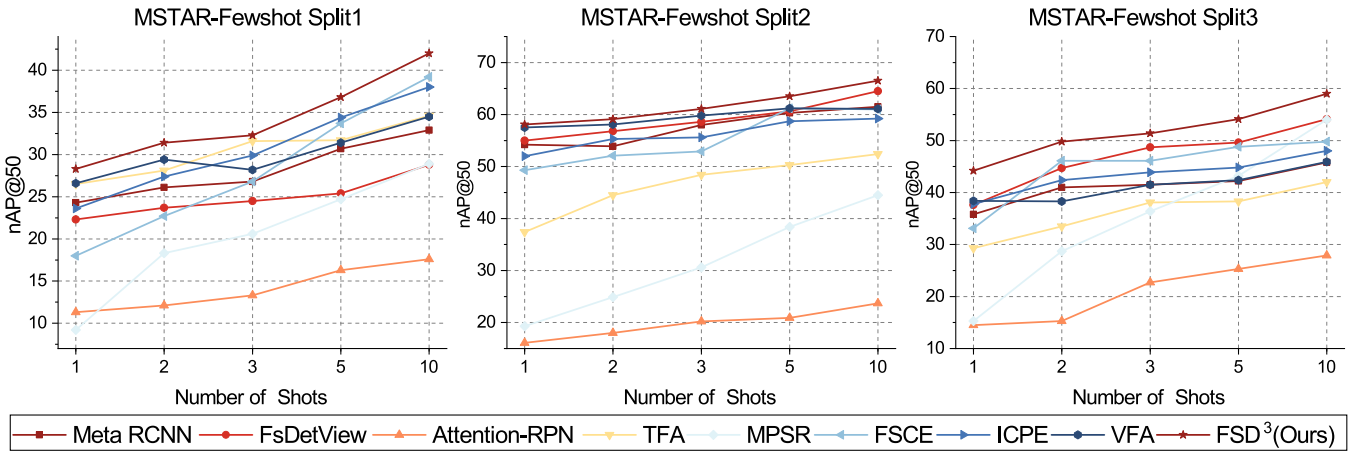


Fig. 7. Experimental results on the MSTAR-Fewshot dataset under multiple runs with different random seeds. It can be observed that the proposed method achieves the best novel class detection performance across all experimental settings.

TABLE II

COMPARATIVE EXPERIMENTAL RESULTS ON THE SAR-AIRCRAFT-1.0 DATASET. RED DENOTES THE OPTIMAL PERFORMANCE, AND BLUE REPRESENTS THE SUBOPTIMAL PERFORMANCE

Method	Backbone	Novel Split 1					Novel Split 2					Novel Split 3					Mean
		1shot	3shot	5shot	10shot	30shot	1shot	3shot	5shot	10shot	30shot	1shot	3shot	5shot	10shot	30shot	
Single run results:																	
Meta RCNN [34]	ResNet 50	8.6	12.5	20.3	25.4	29.4	2.8	6.4	12.8	14.8	14.6	5.8	16.3	17.3	25.0	27.9	15.99
FsDetView [35]	ResNet 50	**11.1**	15.8	20.4	24.1	24.4	4.9	13.6	13.9	17.2	23.8	1.8	13.9	14.1	17.9	22.3	15.94
Attention-RPN [36]	ResNet 50	3.4	13.5	15.8	17.1	34.1	0.5	9.2	12.6	22.7	30.4	0.9	10.0	15.7	19.9	33.9	15.98
TFA [27]	ResNet 50	4.9	9.3	11.3	17.2	20.2	6.2	**16.0**	17.5	17.6	17.1	6.2	15.3	20.8	**34.9**	32.9	16.49
MPSR [28]	ResNet 50	4.6	16.5	22.3	20.1	28.6	1.4	7.5	11.6	15.8	30.5	2.8	15.7	16.9	17.5	33.0	16.32
FSCE [29]	ResNet 50	3.4	12.2	16.4	**25.8**	**36.9**	1.9	**14.8**	16.1	**30.0**	**36.7**	6.1	**17.0**	**21.9**	**35.7**	**44.9**	**21.32**
ICPE [38]	ResNet 50	6.3	12.2	14.2	13.6	21.8	**7.8**	10.1	13.6	13.1	14.2	**6.5**	13.1	19.5	19.3	19.6	13.66
VFA [37]	ResNet 50	10.3	**17.2**	**22.6**	23.8	27.8	4.5	13.9	**18.4**	22.2	25.1	3.3	11.6	9.8	15.8	15.9	16.15
FSD^3(Ours)	ResNet 50	**11.9**	**21.1**	**27.0**	**29.8**	**47.1**	**12.1**	14.7	**19.4**	**27.2**	**42.7**	**15.7**	**23.0**	**23.3**	30.1	**47.5**	**26.17**
Average results over multiple runs:																	
Meta RCNN [34]	ResNet 50	6.4	10.4	13.4	17.4	25.2	4.3	6.7	7.8	10.0	17.7	5.3	10.9	10.7	21.2	26.5	12.93
FsDetView [35]	ResNet 50	6.1	8.5	11.6	17.2	22.9	5.0	8.4	9.5	14.2	24.5	4.4	9.3	10.5	14.3	20.2	12.44
Attention-RPN [36]	ResNet 50	4.1	6.9	9.8	14.4	31.0	3.0	5.7	7.2	15.1	31.8	2.9	7.5	8.7	14.3	27.8	12.68
TFA [27]	ResNet 50	5.1	7.3	10.2	14.7	19.7	**8.8**	**12.8**	**13.6**	16.7	21.4	**7.2**	**12.8**	**17.3**	**24.7**	28.1	14.69
MPSR [28]	ResNet 50	6.2	7.7	10.5	16.5	**35.9**	2.9	5.1	7.1	13.8	31.7	4.7	8.9	8.5	12.1	28.2	13.21
FSCE [29]	ResNet 50	9.1	11.2	17.0	**23.7**	34.8	6.2	8.7	12.0	**21.0**	**36.3**	5.1	8.7	10.6	22.7	**39.6**	**17.78**
ICPE [38]	ResNet 50	7.3	12.9	15.4	16.3	22.6	5.5	8.3	9.5	10.4	13.8	7.4	12.4	17.0	19.4	20.9	13.27
VFA [37]	ResNet 50	**10.8**	**14.2**	**17.4**	21.3	28.0	7.3	10.1	12.3	16.7	27.1	6.1	10.7	10.9	14.5	17.7	15.01
FSD^3(Ours)	ResNet 50	**11.1**	**16.1**	**19.1**	**25.3**	**41.2**	**11.6**	**13.1**	**15.3**	**24.2**	**38.3**	**19.7**	**21.3**	**23.7**	**29.3**	**41.8**	**23.41**

class split 2 and split 3 (−2.8% and −5.6%). In addition, across the 15 different experiments, the proposed method achieves the maximum performance improvement of 10.2% compared to the suboptimal results. Therefore, the proposed method attains the best novel class detection performance under both extreme few-shot conditions (e.g., 1-shot) and relatively sufficient sample conditions (e.g., 30-shot), partially addressing the limited scalability of metalearning-based methods (e.g., Meta RCNN [34], FsDetView [35], VFA [37], and ICPE [38]). Since the proposed method consistently achieves the best novel class detection performance across various settings, it exhibits robustness to different class splits and novel class training samples.

Fig. 6 illustrates the variation of novel class detection performance with the number of shots. It is observed that the proposed method exhibits relatively strong performance even at 1-shot, which is further improved with the increment of shots. Fig. 8 visualizes the detection results for various methods in the SAR-AIRCraft-1.0 [6] dataset under the 30-shot setting for class split 1, where all models adopt the identical fine-tuning samples (i.e., support samples in metalearning) during training and inference. Specifically, the first column represents the ground truth, while the rest columns present the detection results of different methods. It is clear that the proposed method achieves the best performance across diverse scenarios.

2) Result of MSTAR-Fewshot: Table III presents the experimental results on the MSTAR-Fewshot dataset, where red indicates the best novel class detection performance and blue represents the suboptimal performance. It is observed that the proposed method achieves the best novel class detection performance across various settings, significantly outperforming the suboptimal results in some cases with improvements of 4.5% (54.1% versus 49.6%), 3.9% (59.0% versus 54.1%), and

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TABLE III

COMPARATIVE EXPERIMENTAL RESULTS ON THE MSTAR-FEWSHOT DATASET. **RED** DENOTES THE OPTIMAL PERFORMANCE, AND **BLUE** REPRESENTS THE SUBOPTIMAL PERFORMANCE

Method	Backbone	Novel Split 1					Novel Split 2					Novel Split 3					Mean
		1shot	2shot	3shot	5shot	10shot	1shot	2shot	3shot	5shot	10shot	1shot	2shot	3shot	5shot	10shot	
Single run results:																	
Meta RCNN [34]	ResNet 50	**28.8**	28.5	30.5	36.4	37.1	49.0	51.6	56.4	59.2	64.5	35.4	44.9	45.1	45.5	45.4	43.89
FsDetView [35]	ResNet 50	24.8	25.1	25.2	27.7	30.4	54.7	56.5	**59.2**	60.3	**65.9**	33.8	43.9	42.4	44.4	46.7	42.73
Attention-RPN [36]	ResNet 50	9.7	13.8	14.0	16.4	16.6	19.4	19.3	21.8	23.7	23.2	10.4	17.9	21.5	25.4	28.3	18.76
TFA [27]	ResNet 50	18.8	24.6	29.7	30.1	32.6	9.6	47.3	48.6	48.9	50.0	16.9	26.4	29.1	29.5	30.5	31.51
MPSR [28]	ResNet 50	3.1	7.9	7.1	11.1	14.4	18.8	21.8	23.6	22.6	32.4	2.7	21.0	31.0	31.9	38.5	19.19
FSCE [29]	ResNet 50	26.2	30.3	32.9	**36.8**	**41.1**	**58.7**	**59.6**	60.4	**61.3**	63.0	**43.6**	**48.0**	**49.2**	**49.2**	**50.7**	**47.40**
ICPE [38]	ResNet 50	15.0	25.8	24.8	35.0	38.9	54.3	53.4	54.6	55.7	56.4	36.9	43.1	42.5	42.8	46.0	41.68
VFA [37]	ResNet 50	28.6	**32.6**	**34.7**	34.9	35.0	57.0	58.0	58.4	58.6	63.6	31.6	35.0	39.7	40.9	43.4	43.47
FSD^3(Ours)	ResNet 50	**31.5**	**33.2**	**39.1**	**40.2**	**42.8**	**59.9**	**60.7**	**61.5**	**63.4**	**67.2**	**46.4**	**52.0**	**52.6**	**57.0**	**63.8**	**51.42**
Average results over multiple runs:																	
Meta RCNN [34]	ResNet 50	24.3	26.1	26.8	30.7	32.9	54.2	53.9	58.0	60.3	61.5	35.8	41.0	41.5	42.2	45.8	42.33
FsDetView [35]	ResNet 50	22.3	23.7	24.5	25.4	28.8	55.0	56.8	58.6	60.6	**64.5**	37.6	44.7	**48.7**	**49.6**	**54.1**	**43.66**
Attention-RPN [36]	ResNet 50	11.3	12.1	13.3	16.3	17.6	16.1	18.0	20.2	20.9	23.7	14.5	15.3	22.7	25.3	27.9	18.35
TFA [27]	ResNet 50	26.5	28.1	**31.6**	31.7	34.6	37.4	44.5	48.4	50.3	52.4	29.3	33.5	38.1	38.3	42.0	37.78
MPSR [28]	ResNet 50	9.2	18.3	20.6	24.7	28.9	19.3	24.9	30.6	38.4	44.5	15.3	28.7	36.4	43.2	54.0	29.13
FSCE [29]	ResNet 50	18.0	22.7	26.8	33.7	**39.2**	49.3	52.1	52.9	**61.2**	61.0	33.1	**46.1**	46.1	48.8	49.8	42.72
ICPE [38]	ResNet 50	23.6	27.4	29.9	**34.4**	38.0	52.0	55.3	55.6	58.7	59.2	37.8	42.4	43.9	44.8	48.0	43.40
VFA [37]	ResNet 50	**26.6**	**29.4**	28.2	31.4	34.5	**57.5**	**58.1**	**59.8**	**61.2**	61.1	**38.4**	38.3	41.5	42.4	45.9	43.62
FSD^3(Ours)	ResNet 50	**28.3**	**31.4**	**32.3**	**36.8**	**42.0**	**58.1**	**59.1**	**61.1**	**63.5**	**66.5**	**44.2**	**49.8**	**51.4**	**54.1**	**59.0**	**49.17**

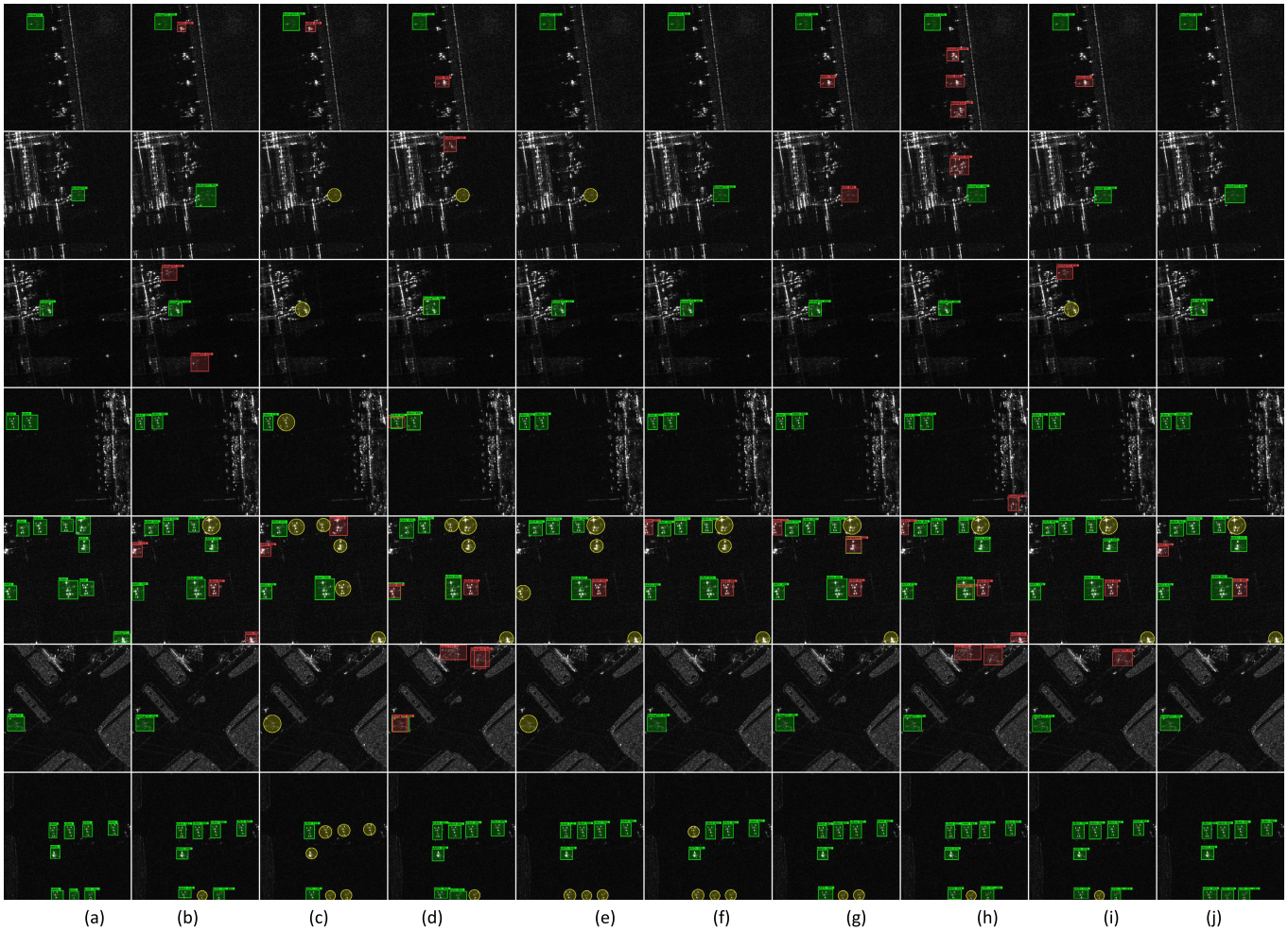


Fig. 8. Visualization of detection results on the SAR-AIRCraft-1.0 dataset. From left to right: (a) ground truth and detection results of (b) Meta RCNN [34], (c) FsDetView [35], (d) Attention RPN [36], (e) TFA [27], (f) MPSR [28], (g) FSCE [29], (h) VFA [37], (i) ICPE [38], and (j) FSD^3 (ours), where the green rectangles indicate correctly detected targets, the red rectangles represent false alarms or misclassifications, and the yellow circles denote missed targets.

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TABLE IV

ABLATION STUDY OF VARIOUS COMPONENTS ON THE SAR-AIRCRAFT-1.0 DATASET ACROSS MULTIPLE RUNS

Method	DP-RPN	CKD	DCDH	Novel Split1					Novel Split2					Novel Split3					Mean
				1-shot	3-shot	5-shot	10-shot	30-shot	1-shot	3-shot	5-shot	10-shot	30-shot	1-shot	3-shot	5-shot	10-shot	30-shot	
Meta RCNN				6.4	10.4	13.4	17.4	25.2	4.3	6.7	7.8	10.0	17.7	5.3	10.9	10.7	21.2	26.5	12.93
FSD^3(Ours)	✓			7.4	11.8	13.2	19.5	32.2	6.2	7.8	9.6	18.9	28.9	10.4	14.0	14.9	22.1	32.3	16.61
	✓	✓		9.7	14.2	15.7	23.7	36.4	8.8	10.5	10.9	21.8	34.2	14.7	18.8	19.7	23.8	36.1	19.87
	✓	✓	✓	11.1	16.1	19.1	25.3	41.2	11.6	13.1	15.3	24.2	38.3	19.7	21.3	23.7	29.3	41.8	23.41
Improved				+4.7	+5.7	+5.7	+7.9	+16.0	+7.3	+6.4	+7.5	+14.2	+20.6	+14.4	+10.4	+13.0	+8.1	+15.3	+10.48

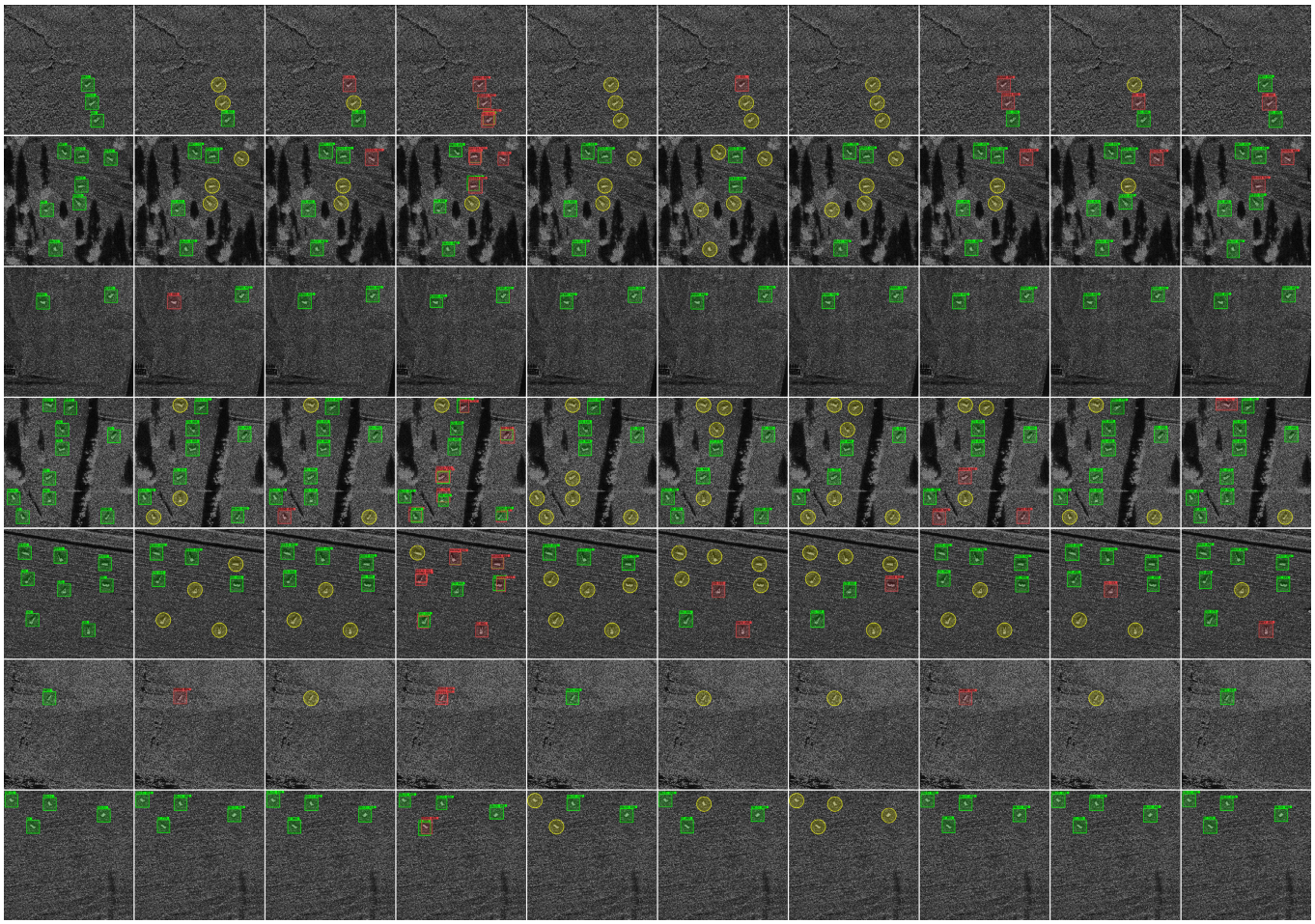


Fig. 9. Visualization of detection results on the MSTAR-Fewshot dataset. From left to right: (a) ground truth and detection results of (b) Meta RCNN [34], (c) FsDetView [35], (d) Attention RPN [36], (e) TFA [27], (f) MPSR [28], (g) FSCE [29], (h) VFA [37], (i) ICPE [38], and (j) FSD^3 (ours), where the green rectangles indicate correctly detected targets, the red rectangles represent false alarms or misclassifications, and the yellow circles denote missed targets.

TABLE V

ABLATION STUDY OF VARIOUS COMPONENTS ON THE MSTAR-FEWSHOT DATASET ACROSS MULTIPLE RUNS

Method	DP-RPN	CKD	DCDH	Novel Split1					Novel Split2					Novel Split3					Mean
				1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot	
Meta RCNN				24.3	26.1	26.8	30.7	32.9	54.2	53.9	58.0	60.3	61.5	35.8	41.0	41.5	42.2	45.8	42.33
FSD^3(Ours)	✓			25.8	28.6	29.2	31.8	36.9	55.4	56.9	60.0	61.1	64.3	39.3	43.4	47.0	48.8	52.1	45.37
	✓	✓		26.8	29.9	31.5	34.3	40.3	56.2	58.7	61.9	62.9	65.8	40.5	44.3	47.8	50.5	53.3	46.98
	✓	✓	✓	28.3	31.4	32.3	36.8	42.0	58.1	59.1	61.1	63.5	66.5	44.2	49.8	51.4	54.1	59.0	49.17
Improved				+4.0	+5.3	+5.5	+6.1	+9.1	+3.9	+5.2	+3.1	+3.2	+5.0	+8.4	+8.8	+9.9	+11.9	+13.2	+6.84

3.1% (36.8% versus 33.7%). Fig. 7 illustrates the variation of novel class detection performance with the number of shots. Fig. 9 further visualizes the detection results under class split 1 and 10-shot settings, where the fine-tuning samples (support samples in metalearning) used during training and inference are identical. It can be observed that the proposed method consistently achieves the best detection performance across various scenarios, effectively covering most ground-truth targets and demonstrating a high recall rate. Since the target slices of different classes are remarkably similar in this dataset,

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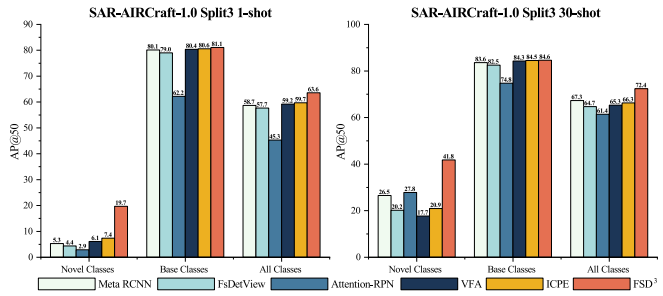


Fig. 10. nAP, bAP, and mAP of various metalearning-based few-shot detection methods on the SAR-AIRCraft-1.0 dataset.

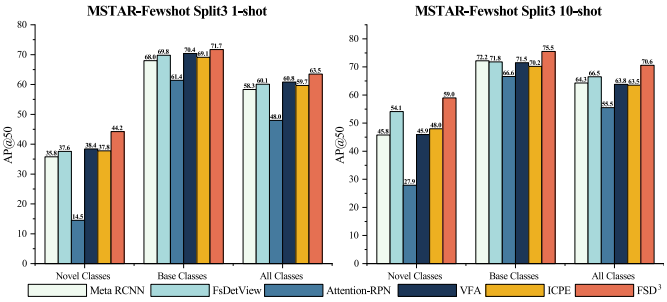


Fig. 11. nAP, bAP, and mAP of various metalearning-based few-shot detection methods on the MSTAR-Fewshot dataset.

it is quite challenging for metric learning-based methods, e.g., Attention-RPN [36], to distinguish them. Therefore, the classifier assigns high confidence scores to multiple classes, leading to a substantial number of false alarms, as demonstrated by the visualization results of Attention-RPN [36].

In addition, Figs. 10 and 11 present the novel class, base class, and overall detection performance of various metalearning methods under 1- and 30-shot conditions for class split 3 of the SAR-AIRCraft-1.0 dataset and the MSTAR-Fewshot dataset. Since transfer-learning methods often employ FPN [59] for multiscale feature extraction, while metalearning methods typically utilize only the C4 layer of the backbone, we consider it unfair to directly compare their detection performance in base classes. Therefore, we only compare the base class performance of different metalearning methods. It is observed that the proposed method not only demonstrates superior performance in novel class detection but also maintains robust performance in base class detection.

E. Ablation Study

1) Evaluation of Different Components: To comprehensively validate the efficacy of the proposed modules, we conducted ablation experiments with Meta RCNN as the baseline. Tables IV and V present results of the ablation experiments for the SAR-Aircraft-1.0 and MSTAR-Fewshot datasets, respectively. It is evident that each module contributes to performance enhancement. Compared to the baseline, joint application of the proposed modules improves the average performance by 10.48% across various experimental settings on the SAR-Aircraft-1.0 dataset, with a maximum improvement of 20.6% and a minimum of 4.7%. On the MSTAR-Fewshot dataset, it increases the average performance by 6.84%, achieving a maximum improvement of 13.2% and a minimum of 3.1%.

Specifically, the replacement of conventional RPN with proposed DP-RPN yields a significant performance enhancement: the DP-RPN improves the average performance by 3.68% on the SAR-Aircraft-1.0 dataset while improving the average performance by 3.04% on the MSTAR-Fewshot dataset. On the basis of DP-RPN, the CKD further enhances the average detection performance for novel classes by 3.26% across different class splits and shot settings on the SAR-Aircraft-1.0 dataset, while improves the average performance by 1.61% on the MSTAR-Fewshot dataset. In addition, it is observed that DCDH improves the average detection performance for novel classes by 3.54% across various experimental settings on the SAR-Aircraft-1.0 dataset and enhances the average performance by 2.19% on the MSTAR-Fewshot dataset.

2) Ablation of DP-RPN: Our proposed DP-RPN dynamically transfers semantic information from support targets to modulate local features in query images through dynamic network construction, effectively mitigating insufficient recall of novel classes caused by background clutter. As manifested in Tables IV and V, replacing the standard RPN in the Meta RCNN baseline with DP-RPN yields substantial performance gains. To further validate the effectiveness of DP-RPN, we compute the AR rates at 5, 10, and 100 retained targets (denoted as AR@5, AR@10, and AR@100) and compare them with those of traditional RPN and unfrozen RPN, respectively. As demonstrated in Tables VI and VII, DP-RPN significantly enhances the recall rates for novel classes under few-shot conditions. In addition, we observe that unfreezing RPN parameters during few-shot fine-tuning partially alleviates its inherent bias toward base-class proposals and improves the detection performance for novel classes. However, such a strategy induces catastrophic forgetting of base classes, consequently degrading their detection performance. In contrast, DP-RPN achieves superior novel-class recall without parameter unfreezing, thereby mitigating catastrophic forgetting of base classes effectively.

3) Ablation of CKD: To validate the effectiveness of the proposed CKD, we conduct ablation studies built upon DP-RPN and provide detailed results in Tables VIII and IX. The CKD module comprises two components: feature encoding, which enhances the discriminability among different classes, and category knowledge decoding, which leverages the query RoI features as prompts to decode the encoded class knowledge, effectively addressing the feature-label misalignment issue after feature aggregation. Tables VIII and IX present the averaged ablation results for the two components on the SAR-Aircraft-1.0 and MSTAR-Fewshot datasets under split 1, respectively, where "Encode" denotes class knowledge encoding and "Decode" denotes class knowledge decoding. We observe that in the MSTAR-Fewshot dataset, directly decoding support features without encoding the category knowledge has limited promotion on the final detection performance, which may be induced by the high feature similarity among classes before encoding. On the contrary, joint adoption

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TABLE VI
COMPARISON OF THE AR RATES FOR NOVEL CLASSES WITH CLASS SPLIT 1 OF THE SAR-AIRCRAFT-1.0 DATASET

AR	1-shot			2-shot			3-shot			5-shot			10-shot		
	RPN	RPN†	DP-RPN	RPN	RPN†	DP-RPN	RPN	RPN†	DP-RPN	RPN	RPN†	DP-RPN	RPN	RPN†	DP-RPN
AR@5	18.70	19.42	21.10	20.80	24.14	26.28	22.00	22.94	27.70	26.28	26.78	29.28	31.28	33.56	34.90
AR@10	21.04	23.28	26.78	23.18	27.78	31.18	23.16	25.34	32.22	27.28	28.84	33.42	31.70	34.28	38.90
AR@100	22.04	24.40	29.46	23.54	28.18	32.74	23.38	25.82	33.52	27.32	29.24	34.96	31.70	34.40	40.10

TABLE VII
COMPARISON OF THE AR RATES FOR NOVEL CLASSES WITH CLASS SPLIT 1 OF THE MSTAR-FEWSHOT DATASET

AR	1-shot			2-shot			3-shot			5-shot			10-shot		
	RPN	RPN†	DP-RPN	RPN	RPN†	DP-RPN	RPN	RPN†	DP-RPN	RPN	RPN†	DP-RPN	RPN	RPN†	DP-RPN
AR@5	55.16	58.34	65.90	56.48	63.44	68.86	58.30	62.94	69.20	62.38	64.42	69.62	63.62	67.94	73.10
AR@10	59.26	63.90	76.04	60.10	69.50	77.02	61.80	68.56	76.30	66.16	69.20	76.90	65.90	71.86	79.28
AR@100	59.56	64.34	76.98	60.16	69.94	77.70	61.92	68.98	77.40	66.20	69.38	77.94	66.04	71.98	79.92

TABLE VIII
ABLATION RESULTS OF CKD ON THE SAR-AIRCRAFT-1.0 DATASET

Encode	Decode	1-shot	3-shot	5-shot	10-shot	30-shot
Baseline		7.4	11.8	13.2	19.5	32.2
	✓	7.5	11.7	13.6	22.4	36.1
✓		7.2	11.7	14.7	23.3	35.4
✓	✓	9.7	14.2	15.7	23.7	36.4

TABLE IX
ABLATION RESULTS OF CKD ON THE MSTAR-FEWSHOT DATASET

Encode	Decode	1-shot	2-shot	3-shot	5-shot	10-shot
Baseline		25.8	28.6	29.2	31.8	36.9
	✓	26.0	26.6	28.8	30.7	35.1
✓		26.2	30.5	30.4	33.7	39.3
✓	✓	26.8	29.9	31.5	34.3	40.3

TABLE X
IMPACT OF MARGIN PARAMETER ϕ ON DETECTION PERFORMANCE UNDER CLASS SPLIT 1 OF THE MSTAR-FEWSHOT DATASET

Parameters	1-shot	2-shot	3-shot	5-shot	10-shot	Mean
Baseline	26.8	29.9	31.5	34.3	40.3	32.56
$\phi = 0.00$	25.4	30.0	31.1	35.5	41.3	32.66
$\phi = 0.05$	25.9	29.2	29.7	35.7	43.4	32.78
$\phi = 0.10$	28.3	31.4	32.3	36.8	42.0	34.16
$\phi = 0.15$	30.0	29.6	31.2	35.7	40.6	33.42

TABLE XI
ABLATION STUDY OF THE CCL

w/o CCL	1-shot	2-shot	3-shot	5-shot	10-shot	Mean
×	29.6	30.5	31.3	35.9	41.8	33.82
✓	28.3	31.4	32.3	36.8	42.0	34.16

of the encoding module and decoding module can improve the detection performance, indicating that feature encoding can provide highly discriminative category knowledge for the decoding module.

4) Ablation of DCDH: DCDH achieves interclass feature separation and intraclass feature compression by bidirectionally constraining features with the large margin cosine loss and the CCL, thereby improving the classification accuracy and robustness under few-shot conditions, as shown in Tables IV and V. To comprehensively evaluate the effectiveness of DCDH, further experiments are conducted under class split 1 of the MSTAR-Fewshot dataset. First, we discuss the impact of the margin parameter ϕ on the detection performance. As shown in Table X, for a smaller ϕ, the detection performance of novel classes significantly improves for large shots; for a larger ϕ, the detection performance of novel classes is notably enhanced for small shots. To reach a balance, we set $\phi = 0.1$. Furthermore, the influence of CCL is evaluated under the same settings. As shown in Table XI, it improves the detection performance for all shots except for a slight decline in the 1-shot scenario. Such decline may be induced by two reasons: 1) only one training sample is sampled for each class in the 1-shot experimental setup, resulting in significantly higher randomness in the results compared to other experimental setups and 2) since there is only one sample per class, the support branch and query branch actually accept the same training sample, making center calibration unnecessary. In addition, the experimental results show that when the number of samples increases to 2 or above, the CCL can effectively improve the few-shot detection accuracy.

V. CONCLUSION

In order to enhance the detection capability in few-shot conditions, this article proposed FSD^3, which mainly includes the DP-RPN, the CKD, and the DCDH modules. Specifically, the DP-RPN enhances the recall of novel targets by assigning global semantics of different classes to local features in SAR images. Furthermore, the CKD adopts the query features as prompts to guide the aggregation process, ensuring alignment between classes and labels. In addition, the large-margin cosine loss and the CCL are designed to increase interclass margins and minimize the distance between query features and support class prototypes, respectively, in the DCDH, thereby achieving robust and high-performance classification under few-shot conditions. Experimental results on the SAR-

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AIRCraft-1.0 dataset demonstrate that the proposed FSD^3 elevates the mean mAP across all experimental configurations from 12.93 to 23.41. In addition, evaluations on the MSTAR-Fewshot dataset reveal that FSD^3 increases the mean mAP under all testing conditions from 42.33 to 49.17, surpassing existing methods by significant margins.

Future work will focus on developing incremental few-shot object detection for SAR imagery that can sequentially assimilate new classes without requiring additional joint retraining.

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